

ARIANA CHIN

University of California

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RESEARCH INTERESTS: Zamolodchikov periodicity, cluster algebras, algebraic combinatorics

ADVISOR: Pavel Galashin

EDUCATION

University of California, Los Angeles

PhD in Mathematics

Expected June 2027

PhD Candidate

May 2025

M.A. in Mathematics

March 2024

Balbes Award - awarded to the top performing female graduate student

2023-2024

Teaching Experience:

- Upper Division: Probability and Statistics, Enumerative Combinatorics, Real Analysis
- Lower Division: Discrete Mathematics, Linear Algebra, Multivariable Calculus

University of California, Berkeley

B.A. in Mathematics, B.A. in Computer Science

May 2022

Magna Cum Laude

GPA: 3.96

- Percy Lionel Davis Award - awarded to a senior or seniors who have demonstrated excellent scholarship in mathematics
- Regents' and Chancellor's Scholar

Languages: Python, Java, Scheme, C, JavaScript, RISC-V, Matlab, LaTeX, SQL, MongoDB, DynamoDB

RESEARCH

“Half-periodicity of Zamolodchikov periodic cluster algebras” [[pdf](#)][[arXiv: 2602.15140](#)] 2026

- Proved the bipartite belt of length $h + h'$ is a maximal green sequence for every Zamolodchikov periodic B-matrix
- Proved for any Zamolodchikov periodic B-matrix, the form at the half-period is a permutation of the cluster variables of order at most two

“Classification of Zamolodchikov periodic cluster algebras” [[pdf](#)][[arXiv: 2510.18031](#)] 2025

- Proved that the Zamolodchikov periodic cluster algebras are in bijection with pairs of commuting (not necessarily reduced or simply-laced) Cartan matrices of finite type
- Classified all such pairs into 29 infinite families and 14 exceptional types in addition to the 6 infinite families and 11 exceptional types in Stembridge's classification
- Showed these pairs are in 1-to-4 correspondence with nonnegative W-cells for products of two dihedral groups, which encode the action of the standard generators on the Kazhdan–Lusztig basis of the associated Iwahori–Hecke algebra

University of Minnesota, Twin Cities - Research Experience for Undergraduates (REU)

2021

Algebra, Combinatorics, Representation Theory

Minneapolis, MN

“F-Polynomial Ratios in the r -Kronecker” [[pdf](#)][[slides](#)]

- Joint with Noah Caplinger, Nyah Davis, and Swapnil Garg
- Explored limiting cluster variables and F-polynomial ratios in the Kronecker cluster algebra

- Presented an alternate proof for the formal power series expansion of the limit in the 2-Kronecker case, and a closed form in the general r-Kronecker case

“Lattice Models and Puzzles for Dual Weak Symmetric Grothendieck Polynomials” [[pdf](#)][[slides](#)]

- Joint with Elisabeth Bullock, Noah Caplinger, Nyah Davis, and Gahl Shemy
- Constructed a lattice model for the dual weak symmetric Grothendieck polynomials, with a negative result for stable symmetric Grothendieck polynomials
- Presented boundary conditions on a lattice model for Littlewood Richardson coefficients for dual weak symmetric Grothendieck polynomials

Ref: <https://www-users.cse.umn.edu/~reiner/REU/REU.html>

CONFERENCES, WORKSHOPS & TALKS

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| • AMS Sectional: Diagrammatics for Cluster Algebras and Physics - Speaker
<i>Arizona State University</i> | November 2026
Tempe, AZ |
| • University of Michigan Combinatorics Seminar - Invited Speaker
<i>University of Michigan, Ann Arbor</i> | October 2026
Ann Arbor, MI |
| • Richard P. Stanley Seminar in Combinatorics - Invited Speaker
<i>Harvard University</i> | October 2026
Boston, MA |
| • IPAM Tutorials: Quantum Topology, Character Varieties - Lecturer
<i>Institute for Pure and Applied Mathematics, UCLA</i> | September 2026
Los Angeles, CA |
| • Formal Power Series and Algebraic Combinatorics (FPSAC) 2026 - Presenter | Seattle, WA |
| • CMND 2026: Algebraic Combinatorics and Applications - Lightning talk | South Bend, IN |
| • Cornell Combinatorics Seminar (2026) - Invited Speaker
<i>Cornell University</i> | April 2026
Ithaca, NY |
| • UIUC Combinatorics Seminar (2026) - Invited Speaker
<i>University of Illinois, Urbana-Champaign</i> | April 2026
Urbana-Champaign, IL |
| • ALGECOM 2026 - Presenter | West Lafayette, IN |
| • UMN Combinatorics Seminar (2026) - Invited Speaker
<i>University of Minnesota, Twin Cities</i> | April 2026
Minneapolis, MN |
| • NCUWM 2026 - Invited Panelist | Lincoln, NE |
| • Summer School on Algebraic Combinatorics (SSAC) 2025 | Seoul, Korea |
| • Formal Power Series and Algebraic Combinatorics (FPSAC) 2025 | Sapporo, Japan |
| • Graduate Student Combinatorics Conference (GSCC) 2025 - Presenter | Los Angeles, CA |
| • Formal Power Series and Algebraic Combinatorics (FPSAC) 2024 | Bochum, Germany |

OTHER INTERESTS

UCLA Women in Mathematics - organizer	2024-2026
UCLA Symphony, UCLA Philharmonia - violin	2022-2026
UC Berkeley Symphony Orchestra - violin	2019-2022
UC Berkeley Chamber Orchestra - violin	2018-2019